
1. EXECUTIVE SUMMARY

This project analyzes a simulated customer support dataset to examine ticket volume, issue patterns, resolution performance, and customer satisfaction. The analysis was conducted to identify operational patterns across the available support data and evaluate areas related to workload, resolution, and customer experience.

The analysis covers 8,470 customer support tickets across different ticket types, products, priorities, channels, statuses, and customer satisfaction ratings. The data was analyzed through a structured workflow using SQL, Excel, and Power BI, allowing the analysis to progress from detailed data exploration to summarized business insights and interactive reporting.

The analysis identified differences in ticket volume across issue types, products, priorities, and support channels. It also examined the distribution of open, closed, and pending tickets, providing visibility into the observed support workload. Resolution availability was analyzed to assess the presence of documented resolutions, while satisfaction analysis identified variations in customer experience across different support dimensions.

Overall, the findings showed that ticket volume alone does not provide a complete view of customer support performance. The analysis found that ticket type, priority, channel, resolution availability, and customer satisfaction provide additional context for evaluating support workload and identifying areas requiring attention.

Based on these findings, the project concludes with business recommendations focused on workload prioritization, resolution tracking, high-volume issue areas, and customer satisfaction monitoring. The recommendations are based on the patterns identified through the analysis and are intended to support more informed operational decision-making.

2. BUSINESS PROBLEM AND OBJECTIVES

Business Problem

The organization under analysis handles customer support issues across different products, ticket types, priorities, and communication channels. The available support data contains information on ticket volume, status, resolution availability, and customer satisfaction, but these dimensions require structured analysis to identify where the support workload is concentrated, which issues generate higher volumes, and where differences in resolution and customer experience are observed.

The absence of a consolidated analytical view makes it difficult to evaluate support workload across multiple dimensions and identify areas requiring greater operational attention. This project addresses the problem by analyzing customer support ticket data across relevant business dimensions and converting the results into structured findings and business insights.

Project Objectives

The analysis was designed to answer the following key business questions:

- How large is the overall customer support workload?
- Which ticket types, products, and issues generate the highest volume?
- How is the workload distributed across different priority levels and support channels?
- What proportion of tickets are closed, open, or awaiting customer response?
- How consistently are customer issues associated with a documented resolution?
- Which areas show differences in customer satisfaction?
- What operational areas should receive greater attention based on the analysis?

Business Goal

The overall goal of the analysis is to provide a structured, data-driven view of customer support operations and identify areas related to workload management, resolution effectiveness, and customer satisfaction that warrant further operational attention.

3. DATASET AND ANALYSIS APPROACH

Dataset overview

The project uses a simulated customer support dataset containing 8,470 tickets. The dataset includes information about customers, products, ticket types, issues, priorities, channels, ticket status, resolution availability, response and resolution details, and customer satisfaction.

Analysis Workflow

The analysis was completed using three primary tools:

SQL — Used to explore the data, calculate key metrics, identify patterns, and perform cross-dimensional analysis.

Excel — Used to organize analytical results into structured tables for comparison and reporting.

Power BI — Used to create interactive dashboards for monitoring support volume, ticket distribution, resolution performance, and customer satisfaction.

Key Analysis Areas

The project focuses on four major areas:

- **Support Volume:** Overall ticket workload and ticket status.
- **Ticket Distribution:** Patterns across products, ticket types, priorities, and channels.
- **Resolution & Response Performance:** Resolution availability, workload patterns, and ticket status.
- **Customer Satisfaction:** Satisfaction patterns across issues, channels, priorities, and ticket types.

The analysis workflow provided a structured progression from dataset exploration and metric calculation to comparative analysis and visual reporting.

4. KEY FINDINGS

The analysis of 8,470 customer support tickets identified several patterns across support workload, resolution tracking, and customer satisfaction.

1. High Open Workload: The analysis found that 2,769 tickets were closed, while 2,819 remained open and 2,881 were pending customer response. The results show that a substantial portion of the ticket volume was not in a closed status at the time of analysis.

2. Similar Ticket Volume Across Major Types: The five major ticket types showed relatively similar volumes. Refund Requests (1,752) and Technical Issues (1,747) recorded the highest ticket volumes, followed by Cancellation Requests, Product Inquiries, and Billing Inquiries.

3. Resolution Tracking Requires Attention: The analysis examined resolution availability to determine whether tickets had a documented resolution. The results provided visibility into the extent to which resolution information was available across the support dataset.

4. Priority and Issue Mix Affect Workload: The analysis found differences in ticket volume across combinations of ticket type, issue, and priority. The results showed that workload patterns varied when multiple support dimensions were considered together rather than evaluating ticket volume alone.

5. Customer Satisfaction Varies Across Segments: The analysis uncovered differences in average customer satisfaction across issues, channels, and priority levels. These variations identified specific support segments where customer experience differed from other segments.

Overall Finding: Overall, the analysis identified differences in workload, ticket status, resolution availability, and customer satisfaction across multiple support dimensions. The findings indicate that evaluating customer support performance requires consideration of multiple dimensions rather than relying on overall ticket volume alone.

5. CUSTOMER SUPPORT PERFORMANCE

The analysis provides a detailed view of customer support workload across ticket status, ticket type, priority, support channel, and resolution availability.

1. Ticket Status: Out of 8,470 tickets, 2,769 were closed, 2,819 were open, and 2,881 were pending customer response. The results show that a substantial portion of the ticket volume was not in a closed status at the time of analysis.

2. Ticket Type: Refund Requests and Technical Issues recorded the highest ticket volumes, with 1,752 and 1,747 tickets respectively. The results identify these categories as the largest contributors to the overall support workload.

3. Priority: The analysis showed differences in ticket volume across priority levels. Combining priority with ticket type provided additional visibility into how workload was distributed across different support categories.

4. Support Channel: The analysis found that tickets were distributed across multiple support channels. Channel-level analysis provided visibility into the distribution of customer demand across the available support channels.

5. Resolution: Resolution availability was analyzed to determine whether tickets contained documented resolution information. The results provided visibility into the availability of resolution data across the support workload and supported further analysis of resolution patterns.

6. Performance Implication: Overall, the analysis identified differences in workload distribution, ticket status, priority, support channel, and resolution availability. These findings provide a basis for evaluating support workload and identifying areas requiring further operational attention.

6. CUSTOMER SATISFACTION

Customer satisfaction analysis was conducted to examine how customer experience varies across different aspects of the support process.

1. Satisfaction by Issue: Average satisfaction was compared across different issue categories to identify variations in customer experience. The analysis highlighted issue categories where average satisfaction differed from other categories, providing visibility into areas requiring further investigation.

2. Satisfaction by Channel: Satisfaction was analyzed across support channels to examine whether customer experience varied depending on how customers contacted support.

3. Satisfaction by Priority: Satisfaction was compared across priority levels to examine whether higher-priority cases were associated with differences in customer experience. The analysis provided visibility into satisfaction patterns across different levels of ticket priority.

Key Insight: The analysis found that customer satisfaction varied across issues, channels, and priority levels. Evaluating these dimensions separately provided additional context beyond the overall satisfaction measure and identified specific segments where customer experience differed.

Business Implication

The results indicate that satisfaction patterns can be evaluated at a segment level to identify areas with comparatively lower customer satisfaction. These variations provide a basis for further investigation into factors such as resolution quality, response handling, and support process gaps.

Overall, segment-level satisfaction analysis provides additional context for evaluating customer experience and identifying areas requiring further operational attention.

7. RECOMMENDATION AND CONCLUSION

Business Recommendations

- 1. Prioritize Unresolved Workload:** Monitor open and pending tickets regularly and establish clear follow-up processes to move tickets toward closure.
- 2. Focus on High-Volume Issues:** Give additional attention to high-volume ticket categories, particularly Refund Requests and Technical Issues, to identify recurring causes and reduce avoidable workload.
- 3. Use Priority for Workload Management:** Combine ticket volume with priority levels when allocating support resources to ensure that higher-priority cases receive appropriate attention.
- 4. Strengthen Resolution Tracking:** Ensure that resolution information is consistently recorded for completed tickets. This will improve reporting consistency and provide a useful knowledge base for identifying recurring problems.
- 5. Monitor Satisfaction by Segment:** Track customer satisfaction across issues, channels, and priority levels rather than relying only on the overall satisfaction score. Segments with comparatively lower satisfaction should be investigated further.

Conclusion

This project demonstrates how customer support data can be transformed into meaningful business insights through SQL, Excel, and Power BI. The analysis identified workload patterns, resolution gaps, and differences in customer satisfaction that can support better operational decision-making.

The key opportunity is to move from simply monitoring ticket volume to actively managing workload, resolution effectiveness, and customer experience using data-driven insights.